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COMMON QUARTERLY EXAMINATION – 2025

Standard XII

Reg No.: | | | | | | |

BIOLOGY

Time: 3.00 hrs. **Part - I (Bio-Botany) 35 marks** **Marks: 70**

I. Choose the correct answer: **8 x 1 = 8**

1. First cell of male gametophyte in angiosperm is _____.
   a) microspore b) megaspore c) nucleus d) Primary endosperm nucleus
2. Ruminate endosperm is found in _____.
   a) Cocos b) Areca c) Vallisneria d) Arachis
3. How many different kinds of gametes will be produced by a plant having the gametophyte AABbCC?
   a) three b) four c) nine d) two
4. Which of the following in the start codon?
   a) AUG b) UGA c) UAA d) UAG
5. Which one of the following palindromic base sequence in DNA can be easily cut at about the middle by particular restriction enzymes?
   a) 5' CGTTCG 3' 3' ATCGTA 5' b) 5' GATATG 3' 3' CTACTA 5'
   c) 5' GAATTC 3' 3' CTTAAG 5' d) 5' CACGTA 3' 3' CTCAGT 5'
6. Match the following.

Column A	*Column B*
1) Totipotency	a) Reversion of mature cells into meristems
2) De differentiation	b) Biochemical and structural changes of cells
3) Explant	c) Properties of living cells develops into entire plant
4) Differentiation	d) Slected plant tissue transferred to culture medium

7. _____ isotopic and non isotopic are popular tools for identification of viruses and other pathogenes.
   a) Bio fuel b) DNA propes c) PCR d) Transfection
8. Solidifying agent used in plant tissue culture is _____.
   a) nicotinic acid b) cobaltous chloride d) EDTA d) Agar

II. Answer any 4 questions. **4 x 2 = 8**

9. What is endothelium?
10. Define the term Diplospory?
11. What is meant by true breeding or pure breeding lines / strain?
12. Name any two physical mutagenes.
13. Write the advantages of Bt cotton.
14. Give the example for micropropagation in plants.

III. Answer any 3 questions. (Q.No.19 is compulsory) **3 x 3 = 9**

15. Distinguish mound layering and air layering.
16. Bring out the inheritance of chloroplast gene with an example.
17. What are the importance of crossing over?
18. Write the advantages of herbicide tolerant plant.
19. What do you mean Embryoids? Write its application.

IV. Answer all the questions. **2 x 5 = 10**

20. a) What is Endosperm? Explain the types. ***(OR)***
    b) Differentiate continuous variation and dis-continuous variation.
21. a) What are the steps involved in recombinant DNA technology? ***(OR)***
    b) Give an account on cryopreservation.

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Part - II (Bio-Zoology) 35 marks

I. Choose the correct answer: **8 x 1 = 8**

1. The foetal membrane that forms the basis of the umbilical cord is
   a) Allantois b) Amnion c) Chorion d) Yolk sac
2. Which of the following are used as genetic markers?
   a) repetitive DNA b) variable number tandem repeats
   c) satellite DNA d) probe DNA
3. B cells that produce and release large amounts of antibody are called
   a) memory cells b) basophils c) plasma cells d) killer cells
4. Father of a child is colourblind and mother is carrier for colourblindness, the probability of the child being colourblind is
   a) 25% b) 50% c) 100% d) 75%
5. MTP was legalized by Government of India in
   a) 1971 b) 1974 c) 1970 d) 1973
6. Choose the mismatched pair.
   a) Tape worm - Apolysis
   b) Noetidure - Endogenous budding
   c) Ceratium - Repeated fission
   d) Amoeba - Encystment
7. The phenomenon of "Industrial Melanism" demonstrates a) Natural selection b) Induced mutation c) reproductive isolation d) Geographical isolation
8. The Athlete's foot disease in human is caused by
   a) Bacteria b) Fungi c) Virus d) Protozoan

II. Answer any 4 questions. **4 x 2 = 8**

9. Mention the symptoms of Phenylketonuria.
10. Differentiate foeticide and infanticide.
11. List out the chemical alarm signals produced during inflammation.
12. Placenta is an endocrine tissue. Justify.
13. Give reasons : "Gnetic code is universal".
14. What do you mean by conjugation? Give example.

III. Answer any 3 questions. (Q.No.19 is compulsory) **3 x 3 = 9**

15. Mention the importance of the position of the testes in humans.
16. Write short notes on withdrawal symptoms.
17. State any three goals of Human genome project.
18. Amniocentesis, the foetal sex determination test, is banned in our country. Is it necessary? Comment.
19. What are called molecular clocks? Mention the molecules which are used to study evolution.

IV. Answer all the questions. **2 x 5 = 10**

20. a) Explain the various phases of menstrual cycle.

(OR)

b) Explain the genetic basis of ABO blood grouping in man.

21. a) Mention the main objections to Darwinism.

(OR)

b) Explain the structure of Immunoglobulin with suitable diagram.

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